Study protocol

Template for an observational study or a simple trial (SPIRIT-style)

Foundations of Clinical Research · Template version 1.0

How to use this template

- Replace every [grey italic text in brackets] with your own text. Press Ctrl + H and search for "[" to step through them, and check none remain before you share the document.
- Grey italic guidance and shaded "Example" boxes are there to help you. Delete them as you write: Home → Styles → right-click "Guidance" (or "Example") → Select All → Delete.
- How this differs from the research proposal template: the proposal is your course product, a plan of 6 to 8 pages for feedback on the idea. The protocol is the full working document of the study: it adds every operational detail (schedule of assessments, data management, monitoring and safety, consent procedures) and is what the ethics committee and the trial registry receive. Write it once the proposal is agreed.
- The structure follows the SPIRIT guidance for trial protocols (www.spirit-statement.org). For an observational study, skip the sections marked "trials only" and describe exposures instead of interventions; use STROBE to check what your final paper will need to report.
- The protocol is the rule book of your study: write it before you start, get it approved, and record every change as a new version.

Administrative information

Item	Details
Full title	*[Descriptive title including the design, e.g. "...: a prospective cohort study"]*
Short title / acronym	*[ACRONYM]*
Protocol version and date	*[Version 1.0]*, *[DD/MM/YYYY]*
Registration	*[Registry name and number (required for trials before the first patient; recommended for observational studies)]*
Sponsor	*[Institution responsible for the study]*
Funding	*[Funder and grant number, or "unfunded / departmental resources"]*
Principal investigator	*[Name, profession, department, email, phone]*
Co-investigators and roles	*[Name and role in the study]*
Study site(s)	*[Department, Institution]*
Ethics committee	*[Ethics committee name]*; approval no. *[number]* of *[DD/MM/YYYY]*

Version history

Version	Date	Main changes	Approved by ethics committee on
[1.0]	*[DD/MM/YYYY]*	Original version	*[DD/MM/YYYY]*
[1.1]	*[DD/MM/YYYY]*	*[Describe the amendment]*	*[DD/MM/YYYY]*

Signature of the principal investigator

By signing, the principal investigator agrees to conduct the study according to this protocol, Good Clinical Practice, the Declaration of Helsinki and [National regulations].

Name: *[Name]*

Signature: _____

Date: *[DD/MM/YYYY]*

Abbreviations

Abbreviation	Meaning
CRF	Case report form
ICU	Intensive care unit
SOFA	Sequential Organ Failure Assessment
[ABC]	*[Meaning]*

1 Synopsis

A one-page summary of the whole study. Fill it in last.

Item	Summary
Title	*[Full title]*
Design	*[e.g. prospective cohort; pilot randomised controlled trial]*
Setting	*[Department, Institution; number of sites]*
Population	*[Main inclusion and exclusion criteria]*
Intervention / exposure	*[What]*
Comparator	*[Compared with what]*
Primary outcome	*[Outcome, measure, time point]*
Secondary outcomes	*[List]*
Sample size	*[N, with one-line justification]*
Duration	*[Recruitment period; follow-up per participant; total study duration]*

2 Background and rationale

What is the problem and how big is it? What is known (cite the key studies, and any systematic review)? What is the gap? Why is this study needed now, and why in this setting? 1–2 pages.

[Write here]

3 Objectives and hypotheses

One primary objective (linked to the primary outcome) and a few secondary objectives. State the hypothesis for the primary objective.

> **Example:** *Primary objective: to estimate the association between cumulative fluid balance (ml/kg) at day 5 and 28-day mortality in adults with septic shock, adjusted for illness severity.*

[Write here]

4 Study design

Name the design (e.g. prospective cohort, parallel-group pilot randomised trial) and explain why it suits the question. For trials: allocation ratio and framework (superiority, non-inferiority). A simple flow diagram helps.

[Write here]

5 Participants

5.1 Setting

Type of unit and hospital, number of beds, patients seen per year, study period.

[Write here]

5.2 Eligibility criteria

Inclusion criteria	Exclusion criteria
[Inclusion 1]	*[Exclusion 1]*
[Inclusion 2]	*[Exclusion 2]*
[Inclusion 3]	*[Exclusion 3]*

5.3 Screening and recruitment

Who screens, when and how? Keep a screening log (screened, eligible, included, reasons for non-inclusion). You will need it for the CONSORT or STROBE flow diagram. State the expected recruitment rate and what you will do if recruitment is slow.

[Write here]

6 Interventions (trials) or exposures (observational studies)

Trials: describe the intervention and comparator in enough detail to be repeated (what, who, how much, how often, how long), criteria for stopping or changing it, how adherence is checked, and which other treatments are allowed. Observational studies: define the exposure precisely and how and when it is measured.

[Write here]

7 Outcomes

For each outcome: the variable, how it is measured, the unit, and the time point. Only ONE primary outcome.

Type	Outcome	Definition and measurement	Time point
Primary	*[Outcome]*	*[Definition]*	*[e.g. Day 28]*
Secondary	*[Outcome]*	*[Definition]*	*[Time point]*
Secondary	*[Outcome]*	*[Definition]*	*[Time point]*
Safety (trials)	*[Adverse event]*	*[Definition]*	*[Time point]*

8 Participant timeline (schedule of assessments)

Mark with X what is done at each time point. Adapt the columns to your study.

Assessment	Screening	Enrolment (Day 0)	Days 1–5 (daily)	Day 7	ICU discharge	Day 28
Eligibility check	X					
Consent / information	X	X				
Demographics and medical history		X				
Severity score (e.g. SOFA)		X				
Laboratory values (e.g. lactate)		X	X			
Fluid input and output		X	X			
Adverse events (trials)			X	X	X	
Vital status					X	X
[Assessment]						

9 Sample size

State the primary outcome, the expected value in the comparison group, the smallest difference worth detecting, power (usually 80–90%), significance level (usually 5%), the resulting number, and the allowance for loss to follow-up. Give the source of each assumption. For a pilot or descriptive study, justify the number on feasibility or precision instead.

[Write here]

10 Allocation and blinding (trials only)

Sequence generation (e.g. computer-generated, blocks, stratified by site); allocation concealment (e.g. central web randomisation or sealed opaque envelopes); who enrols and who assigns; who is blinded (patients, clinicians, outcome assessors, statistician) and how unblinding is done in an emergency.

[Write here]

11 Data collection and management

Which CRF and data dictionary (attach as appendices); who collects data, from which source documents, and when; training; how data are entered (e.g. electronic data capture with range checks, or double entry); data cleaning and queries; secure storage, back-up and how long data are kept.

[Write here]

12 Statistical methods

Write the plan BEFORE seeing the data. Keep it simple and specific.

Topic	Plan
Analysis population	*[e.g. all enrolled participants (intention-to-treat for trials)]*
Descriptive statistics	*[mean (SD) or median (IQR); n (%)]*
Primary analysis	*[test or model, effect measure with 95% CI, adjustment variables]*
Secondary analyses	*[For each secondary outcome]*
Missing data	*[How you will prevent it and how you will report and handle it]*
Subgroup / sensitivity analyses	*[Pre-specified only]*
Software	*[e.g. R, Stata, SPSS]*

13 Monitoring and safety

Who checks the quality of the study (monitoring visits, source-data verification of a sample)? For trials: definitions of adverse events (AE) and serious adverse events (SAE), how and how fast they are reported (e.g. SAEs within 24 h to the sponsor and as required to the ethics committee), whether a data monitoring committee is needed, stopping rules and interim analyses. Observational studies: how protocol deviations and data breaches are recorded.

[Write here]

14 Ethics and consent

Ethics committee approval before any participant is included; compliance with the Declaration of Helsinki, Good Clinical Practice and [National regulations]. Who takes consent, how, and with which documents (attach the information sheet and consent form). Vulnerable participants: for patients who cannot consent (sedation, shock, delirium), describe consent from a legal representative, deferred consent or a waiver, and why it is justified. How amendments are submitted.

[Write here]

15 Confidentiality and data protection

Participants identified only by study ID on CRFs and in the database; the key linking ID to identity kept separately and securely by the principal investigator; access limited to named people; compliance with [National data-protection law]; what is shared and with whom.

[Write here]

16 Dissemination, authorship and data sharing

Target journal(s) and reporting guideline (CONSORT, STROBE...); feedback to staff and, where possible, participants; authorship rules (ICMJE criteria); whether a de-identified dataset will be shared and how.

[Write here]

17 Funding and conflicts of interest

Source of funding and the funder's role; any financial or other interests of the investigators.

[Write here]

18 References

NLM/Vancouver style, numbered in the order cited.

1. *[Authors. Title. Journal. Year;Volume:Pages. doi]*
2. *[Authors. Title. Journal. Year;Volume:Pages. doi]*
3. *[Authors. Title. Journal. Year;Volume:Pages. doi]*

19 Appendices

- Appendix 1: Participant information sheet and consent form(s)
- Appendix 2: Case report form (CRF)
- Appendix 3: Data dictionary
- Appendix 4: Budget
- *[Appendix 5: Other, e.g. screening log, letters of support]*